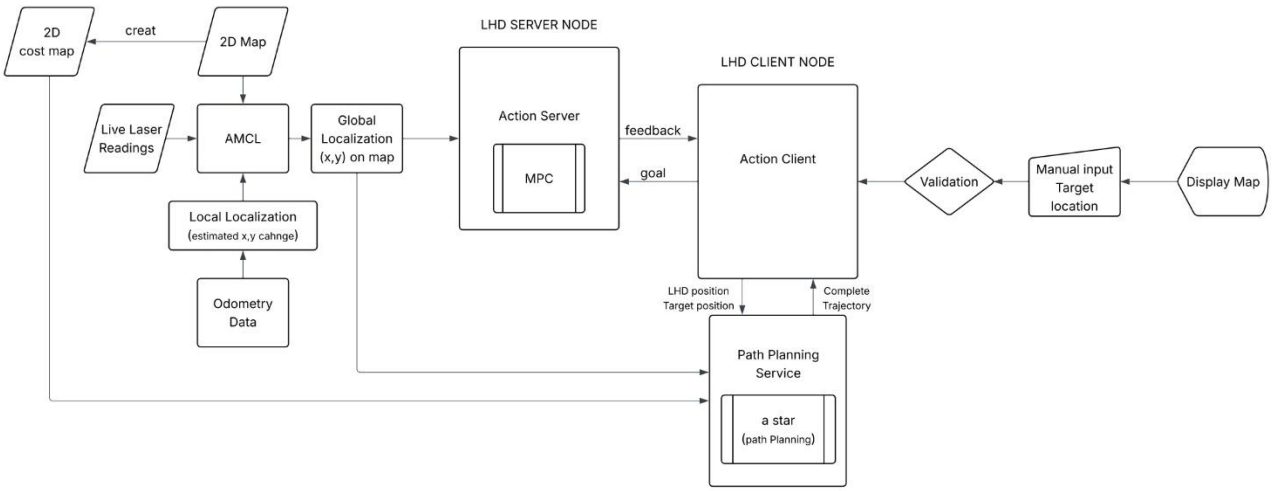
CONTROL SYSTEM ANALYSIS

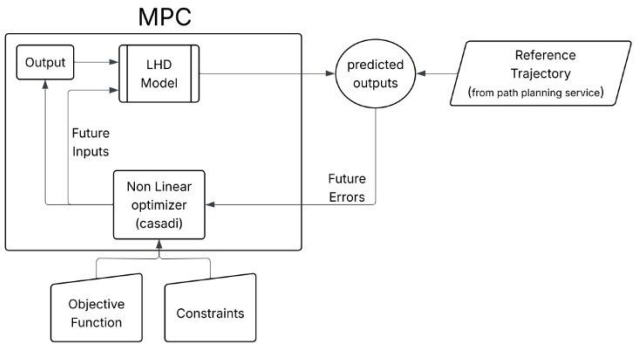
Robotic Load-Haul-Dumper (LHD) scaled system for Mining Automation Technology Development

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1. Introduction

The present invention relates to an autonomous robotic Load-Haul-Dumper (LHD) scaled unit that is configured to replicate the operational characteristics of a full-scale underground LHD machine while providing a modular and cost-effective platform for optimisation, experimentation and demonstration with dimensional scalability. This system consists of scaled vehicular chassis along with a load handling system which is integrated by the electronic control system. The chassis consists of a mix between two-dimensional modular metal components which can easily be CNC machined along with other three-dimensional wedges and connectors that easily be manufactured by additive or conventional manufacturing methods depending upon availability. The modular construction of the model allows for easy customisation of components for optimization.

A 1.5 m x 0.75 m laboratory-scale, centre-articulated Load-Haul-Dump machine scaled unit together with a ROS-based autonomy stack running on a Jetson Orin Nano: AMCL localisation on a 2D occupancy grid, A* global path planning over an inflated costmap, and a nonlinear Model Predictive Controller solved with CasADi that tracks the planned trajectory. A physical prototype has been fabricated; autonomy has been demonstrated in Gazebo simulation on a static map only.

2. Overview

An LHD is the primary mucking machine of underground mining: it loads broken rock at the face, hauls it along a drive, and dumps it at a tip or ore pass. The machine is centre-articulated - two chassis halves joined by a vertical pivot, steered by a pair of opposed hydraulic cylinders acting across that pivot - because articulation gives a tighter turning circle in a confined drive than Ackermann steering can.

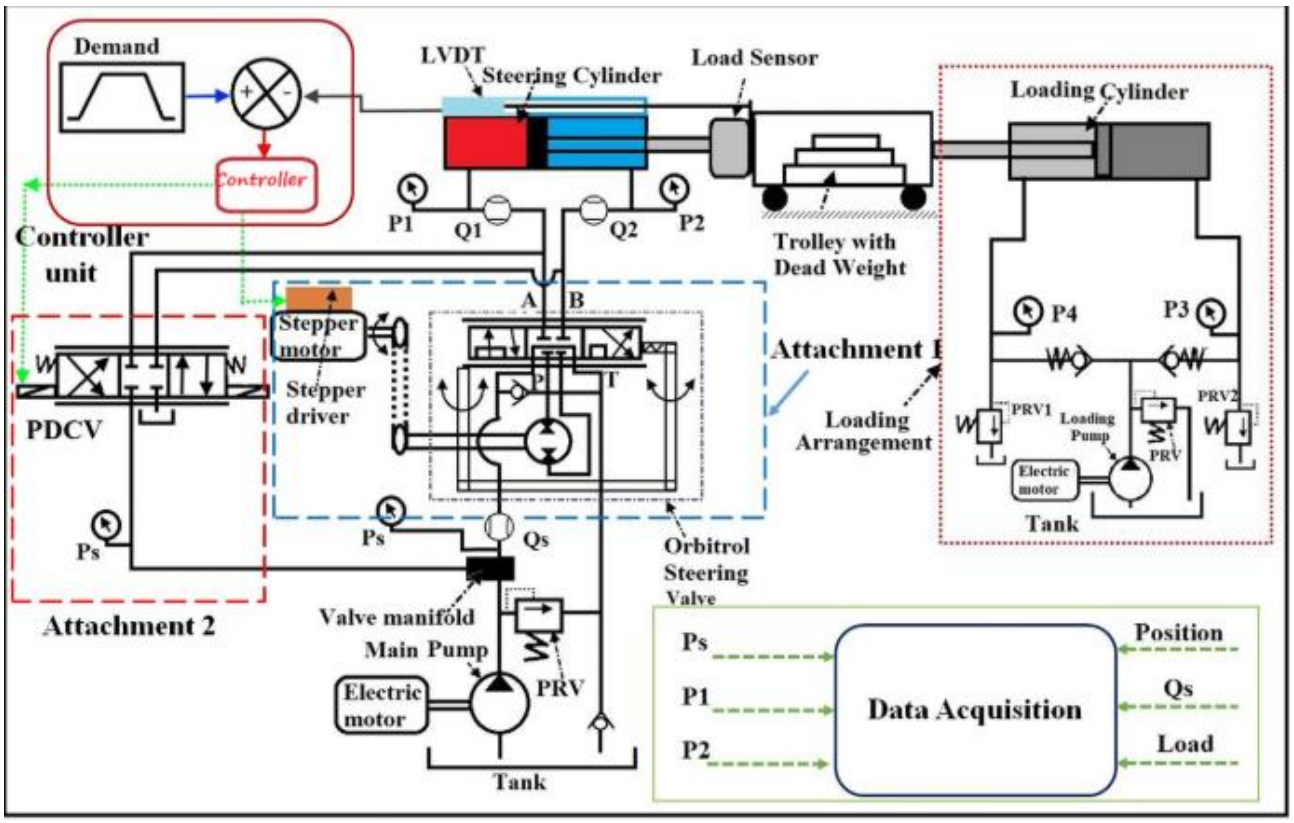
The problem is that manual operation carries 2 to 2.5 hours of idle time per shift, operator fatigue and exposure risk, and poor efficiency in confined ground; and that in the Indian context underground production is a small share of coal output, so indigenous automation of this machine class is strategically motivated. The proposed answer is a low-cost, lab-scale, fully instrumented autonomous LHD serving both as a research platform and as a training rig.

Stated configuration

Attribute	As disclosed	Comment
Overall envelope	1.5 m length, 0.75 m height	Lab scale. Must not appear as a claim limitation; the invention should be scale-independent.
Chassis	Four-wheel, two-section centre-articulated, front section carrying boom and bucket	Conventional LHD topology.
Steering	Opposed hydraulic cylinders across the articulation joint, fed by an orbitrol steering valve	Conventional except for the actuation of the orbitrol - see 3.3.
Front-end	Lift cylinders and bucket linkage	Bucket tilt actuation is not described in the deck. Gap G-04.
Traction	Hydrostatic - main pump drives traction motor	Motor type, displacement and drive topology not disclosed. Gap G-02.
Prime mover	Electric motor driving the main pump	Untethered operation, energy storage and duty rating are not addressed.
Compute	Jetson Orin Nano	Deck slide 4 says 'Jetson Nano', slide 11 says 'Jetson Orion Nano'. Inconsistent - resolve.
Demonstrated state	Physical prototype fabricated (Fig. 10); autonomy demonstrated in Gazebo/RViz simulation only	No hardware-in-the-loop or on-machine autonomy is evidenced. Bears on enablement - Section 7.

3. System architecture as disclosed

The deck presents the system across four figures: a simplified hydraulic circuit of the articulated steering actuation a main control-algorithm flowchart, MPC and A* sub-flowcharts, and the fabricated machine.



3.1 Hydraulic power and actuation subsystem

- **Main circuit.** Electric motor driving the main pump, discharging through a pressure relief valve and a valve manifold to supply line Ps. Supply pressure Ps and supply flow Qs are both instrumented at the manifold outlet.
- **Load branches.** The main pump is stated to drive the traction motor, the lift cylinders and the steering cylinders. The circuit figure details only the steering branch; the traction and lift branches are not drawn.
- **Steering branch.** Two parallel actuation paths are shown into the steering cylinder: an orbitrol steering valve driven at its input shaft by a stepper motor via a stepper driver (the primary path, shown as Attachment 1), and a proportional directional control valve, PDCV, driven from the controller unit (Attachment 2). The two attachments appear to be alternative actuation schemes for comparative evaluation rather than a redundant pair - this must be clarified, as it materially affects how a claim is framed. Gap G-01.
- **Loading arrangement.** A separate loading circuit with its own electric motor and pump, two pressure relief valves (PRV1, PRV2), check valves, and a loading cylinder acting through a trolley carrying a dead weight. This is a test rig that applies a controlled resistive load to the steering actuator - it emulates ground reaction at the articulation joint. It is test apparatus, not part of the machine, and should be described as such.

3.2 Instrumentation

Measurand	Sensor / tag	Function in the loop
Supply pressure	Ps	Circuit supply state; candidate MPC constraint input (Concept B).
Cylinder port pressures	P1, P2	Differential across the steering actuator - load estimate, and basis for a force or torque observer.
Cylinder port flows	Q1, Q2	Actuator velocity cross-check; orbitrol metering characterisation.

Supply flow	Qs	Available flow - the key term if flow availability is to constrain the optimiser.
Rod position	LVDT on steering cylinder	Closes the loop on the open-loop stepper command. This is the element that makes the actuation chain a servo rather than an open-loop indexer.
Reaction load	Load sensor in line with trolley	Rig instrumentation for the applied disturbance.
Loading circuit pressures	P3, P4	Rig instrumentation.
Environment	LiDAR, camera	Scan matching for AMCL, costmap population, obstacle detection.
Acquisition	Central DAQ collecting Ps, P1, P2, Qs, position, load	Sampling rate and synchronisation not stated. Gap G-06.

3.3 Software and control architecture

Figure shows a ROS action-based architecture split into an LHD server node and an LHD client node, with a separate path-planning service.

Element	Role as disclosed
Map and costmap	A 2D occupancy map is modified into a 2D costmap (inflation of obstacles). Map resolution 0.05 m, 539 x 380 cells, per the RViz panel in Fig. 6.
Local localisation	Odometry data yields an estimated incremental pose change.
Global localisation	AMCL fuses live laser readings with the odometry-derived estimate against the map to produce a global (x, y) pose, with a particle cloud shown in Fig. 6.
Operator interface	Target location is entered manually against a displayed map and passed through a validation gate; on failure the operator is returned to re-enter the target. The deck also states target sites can be set autonomously.
Action client	Holds the validated goal, submits LHD position and target position to the path-planning service, receives the complete trajectory, sends the goal to the action server and consumes feedback.
Path-planning service	A* over the costmap. Per Fig. 8b: initialise open_set, came_from, g_score and f_score; expand 8-connected neighbours; bounds check; pixel-cost check; reject impassable (black) pixels; compute tentative g_score; retain if better; reconstruct path by back-tracing from goal to start.
Action server	Runs the MPC against the reference trajectory and returns feedback to the client.
MPC	Per Fig. 8a: an LHD plant model produces predicted outputs; these are differenced against the reference trajectory from the planning service to give future errors; a nonlinear optimiser (CasADi) takes those errors together with an objective function and a constraint set and produces future inputs, which are fed back into the model and applied as the output.
Safety	Emergency stop and dynamic obstacle avoidance are stated as features; dynamic obstacle handling also appears under future work, so the current implementation is static-map only.

4. Description of operation

Reconstructed nominal cycle, stated in the sequence a specification would use:

Step	Action
1	The machine is powered; the electric motor drives the main pump; supply pressure is established at Ps and regulated by the main relief valve.
2	The LiDAR streams scans; AMCL initialises a particle distribution over the prior 2D map and converges to a global pose estimate, aided by wheel and articulation odometry.
3	A target location is supplied, either manually against the displayed map or autonomously by the target-site server. The target is validated as being within bounds and on a traversable cell; an invalid target returns the operator to re-entry.
4	The path-planning service runs A* from the current pose to the target over the inflated costmap and returns a complete geometric trajectory.
5	The action client transmits the goal to the action server. The MPC begins receding-horizon optimisation: at each interval the LHD model predicts outputs over the prediction horizon, the error against the reference is formed, and CasADi solves the constrained nonlinear program for the input sequence.
6	The first element of the optimal input sequence is applied. The articulation component is converted to a target angular displacement at the orbitrol input shaft and issued as a step count and rate to the stepper driver.
7	The orbitrol meters oil to the appropriate steering cylinder port; the cylinders articulate the chassis; the LVDT reports actual rod displacement; residual error against the commanded displacement is corrected on the next cycle.
8	Pose estimate, articulation state and hydraulic state are fed back; the horizon advances and step 5 repeats until the goal tolerance is met.
9	At the face or tip, the lift cylinders actuate the boom for the load or dump phase. (The deck does not describe autonomous bucket loading - the loading cycle appears to remain outside the autonomy scope. Gap G-04.)
10	Emergency stop de-energises the actuation path at any point. The fail-safe state of the steering circuit on power loss is not specified. Gap G-08 - this is a safety-critical omission and also a dependent-claim opportunity.